PHISHING EMAIL DETECTION AND AWARENESS REPORT

Future Interns Cyber Security Internship

Task 2: Phishing Email Detection & Awareness System

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1. Introduction

Email is one of the most widely used communication methods for personal and professional purposes. However, cybercriminals frequently exploit email to distribute spam, conduct phishing attacks, steal sensitive information, and manipulate recipients into clicking malicious links or opening harmful attachments.

Phishing is a type of social engineering attack in which an attacker impersonates a trustworthy person or organization to trick recipients into revealing confidential information or performing unsafe actions.

This project focuses on analyzing email samples, identifying suspicious characteristics, distinguishing legitimate messages from potentially malicious ones, and recommending measures to improve email security awareness.

2. Objectives

The main objectives of this project are:

- Analyze email samples to identify potentially suspicious characteristics.
- Identify indicators such as unsolicited promotions, suspicious links, misleading claims, and unusual requests.
- Classify emails based on the available evidence.
- Assess the potential risks associated with suspicious emails.
- Recommend preventive measures for individuals and organizations.
- Prepare an email security awareness report to help users recognize and avoid email-based threats.

3. Tools and Technologies

The following resources were used in this project:

- **Kaggle:** Source of the email dataset.
- **Microsoft Excel:** Recording email samples, dataset labels, indicators, classifications, risk levels, and recommended actions.
- **Manual text analysis:** Examining email content for suspicious characteristics.
- **Google Docs or Microsoft Word:** Preparing the final report.
- **GitHub:** Maintaining and submitting project files.

4. Dataset Description

The project uses a publicly available email dataset obtained from Kaggle.

Dataset source: https://www.kaggle.com/datasets/naserabdullahalam/phishing-email-dataset

The dataset contains email text and numerical labels. The samples include business correspondence, promotional messages, and automated email notifications.

The original dataset labels were recorded separately from the analyst's assessment. The numerical label mapping should be verified using the dataset documentation before interpreting the labels as legitimate or phishing.

5. Methodology

The analysis followed these steps:

1. **Data collection:** Obtain email samples from the public dataset.
2. **Content examination:** Read each email and identify its apparent purpose.

3. **Indicator identification:** Record observable characteristics such as unsolicited advertising, suspicious requests, unusual claims, and external links.
4. **Classification:** Assign a preliminary classification based on available evidence.
5. **Risk assessment:** Estimate the potential risk associated with the observed characteristics.
6. **Recommendation:** Record appropriate actions for recipients.
7. **Documentation:** Store the results in an Excel workbook and prepare the report.

The analysis is primarily based on the available email text. Where full headers, link destinations, or sender verification were unavailable, classifications remain preliminary.

6. Email Analysis and Findings

Example 1: Unsolicited Product Promotion

Description: An email advertising notebook accessories, batteries, chargers, and discounted computer products.

Observed indicators:

- Bulk commercial advertising.
- Numerous product offers.
- External website addresses.
- Clearance-sale claims and promotional language.

Assessment: The message appears to be unsolicited commercial advertising. The available text does not independently establish that it is phishing.

Classification: Spam / Potentially Suspicious
Preliminary risk: Medium

Recommended action: Verify the seller independently and avoid unverified purchase links.

Example 2: Business Workshop Correspondence

Description: An email discussing workshop arrangements, registered delegates, meeting times, and contract-related matters.

Observed indicators:

- Specific meeting arrangements.
- Relevant business context.
- A named recipient and professional sign-off.
- Discussion of work-related matters.

Assessment: The message appears consistent with ordinary business correspondence. Its authenticity cannot be confirmed from the text alone.

Classification: Safe-looking / Legitimate
Preliminary risk: Low

Recommended action: Verify the sender before acting on confidential or financial requests.

Example 3: Recruitment Recommendation

Description: An internal business email recommending that a company expand recruitment to include mathematics majors.

Observed indicators:

- A coherent business argument.
- Specific recruitment-related details.
- Relevant university and workplace context.
- A professional sign-off.

Assessment: The message appears to be ordinary business correspondence, with no clear phishing indicators in the supplied text.

Classification: Safe-looking / Legitimate
Preliminary risk: Low

Recommended action: Verify sender identity before sharing sensitive information.

Example 4: Delivery Failure Notification Containing Medication Advertising

Description: A returned email includes an original message promoting low-priced medication. The provided headers also contain a spam-filter rejection.

Observed indicators:

- Unsolicited medication promotion.
- Unusually low advertised prices.
- Sexual-performance claims.
- Evidence of spam-filter rejection.

Assessment: The original message has strong indicators of spam. However, spam does not necessarily mean that the message is a phishing attack.

Classification: Spam / Phishing-Related
Preliminary risk: High for the suspicious promotional content

Recommended action: Do not follow promotional links or purchase medication through the email. Report or delete suspicious messages.

Example 5: Automated Delivery Failure Notification

Description: A notification reports that delivery failed because the recipient's mailbox was unavailable or the recipient address was unknown.

Observed indicators:

- SMTP error code 550.
- Permanent delivery failure.

- Recipient reported as unknown or unavailable.

Assessment: These characteristics are consistent with a legitimate bounced email. The text does not provide clear evidence of phishing.

Classification: Delivery Failure Notification / Unverified
Preliminary risk: Low

Recommended action: Check the recipient address if the original email was sent by the user. Verify unexpected notifications before opening attachments.

7. Common Email Security Warning Signs

1. **Unusual sender address:** The sender's address differs from the organization they claim to represent.
2. **Urgent requests:** The message pressures the recipient to act immediately.
3. **Credential requests:** The email asks for passwords, one-time passwords, or sensitive account information.
4. **Suspicious links:** The displayed link does not match the expected destination.
5. **Unexpected attachments:** The email contains an unfamiliar or unexpected file.
6. **Unrealistic offers:** The message advertises unusually cheap products, prizes, or benefits.
7. **Impersonation:** The sender pretends to be a trusted organization or colleague.
8. **Unexpected financial requests:** The email asks for money transfers or payment-detail changes.

No single indicator proves that an email is malicious. Multiple indicators and supporting evidence should be considered together.

8. Potential Risks and Impact

Successful phishing attacks can have serious consequences.

- **Credential theft:** Attackers may obtain passwords and compromise accounts.
- **Financial loss:** Victims may make fraudulent payments or disclose banking information.
- **Malware infection:** Malicious attachments or downloads may compromise a device.
- **Data breaches:** Confidential personal or organizational information may be exposed.
- **Business disruption:** Compromised accounts can be used to target colleagues and customers.
- **Reputational damage:** Organizations may lose customer trust following a security incident.

The actual impact depends on the attack method, the information exposed, and whether the recipient interacts with malicious content.

9. Prevention and Awareness Guidelines

1. Verify the sender's complete email address.
2. Avoid clicking unexpected links; visit the official website independently.
3. Never disclose passwords or one-time passwords in response to unsolicited requests.
4. Verify payment changes through a separate trusted communication channel.
5. Avoid opening unexpected attachments.
6. Keep operating systems, browsers, and security software updated.
7. Enable multi-factor authentication where available.

8. Report suspicious messages to the organization's IT or security team.
9. Use email filtering and security controls to reduce spam and malicious messages.
10. Provide regular security awareness training.

These measures can reduce the likelihood of successful phishing attacks and help users respond safely to suspicious messages.

10. Limitations

This project has several limitations:

- The assessment is based mainly on the email text available in the dataset.
- Sender authenticity was not independently verified for every sample.
- Link destinations and attachment contents were not necessarily available for inspection.
- The dataset's numerical label mapping must be verified before using the labels as ground truth.
- Some samples may represent spam or delivery failures rather than phishing.
- Risk levels are preliminary assessments, not results from a validated automated detection model.

These limitations mean that classifications should be treated as initial assessments rather than definitive security verdicts.

11. Conclusion

This project demonstrates the importance of examining email content, identifying suspicious characteristics, and applying appropriate security precautions.

The analyzed examples show that phishing detection requires careful judgment. Promotional content, delivery failure notices, and legitimate business

correspondence must be evaluated according to their context and the available evidence.

Combining sender verification, safe link-handling practices, email filtering, and user awareness can help reduce the risk of email-based attacks.

The project also highlights the importance of distinguishing between spam and phishing and validating dataset labels before drawing conclusions.

12. References

1. Kaggle, *Phishing Email Dataset*. https://www.kaggle.com/datasets/naserabdullahalam/phishing-email-dataset
2. Federal Trade Commission, *How to Recognize and Avoid Phishing Scams*. https://consumer.ftc.gov/articles/how-recognize-and-avoid-phishing-scams
3. Cybersecurity and Infrastructure Security Agency (CISA), *Recognize and Report Phishing*. https://www.cisa.gov/secure-our-world/recognize-and-report-phishing

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End of Report